Harishnarayanan R

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Bachelor of Engineering
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Portfolio Website | GitHub Profile
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Education

•Rajalakshmi Engineering College, Chennai *2021-2025*
Bachelor of Engineering in Electronics and Communication CGPA: 8.1

•Neelan Matriculation Higher Secondary School *2021*
State Board of Tamilnadu Percentage: 95.6

•Neelan Matriculation Higher Secondary School *2019*
State Board of Tamilnadu Percentage: 92.4

Experience

•Internship Trainee, Southern Railways EMU Workshop *Jun 2023*
Tambaram, Chennai *Certificate*

- Supported operation and maintenance of a 5000 LPH transformer oil filtration system; performed quality tests to assess dielectric strength and purity of oil.
- Gained hands-on experience in electrical insulation techniques, ensuring safe removal of moisture, gases, and particulates while maintaining compliance with industry safety standards.

•Internship Trainee, NSIC - Technical Services Centre *Jan 2023*
Guindy Industrial Estate, Chennai *Certificate*

- Built AI-enabled embedded systems with IoT for smart monitoring and automation in industrial setups.
- Deployed real-time logic modules and data flow systems on embedded hardware to support faster and smarter industrial automation workflows.

Projects

-Airbag System for Elderly Fall Protection with Proactive Safety Measures *Project Report*

- Engineered an Arduino-based embedded fall-detection system using a MEMS accelerometer and gyroscope, smartly integrating a wearable inflatable airbag to automatically protect elderly users during falls.
- Developed and fine-tuned a threshold-based fall detection algorithm using Arduino and MEMS sensors, enabling real-time identification of falls and triggering airbag activation to help prevent injuries.
- Incorporated biometric health-monitoring sensors and real-time alerts into the wearable device to improve safety and overall care for elderly users.

-Efficient Design of Wearable Antenna for Biomedical Applications *Project Report*

- Designed and optimized a compact microstrip patch antenna tailored for wearable biomedical applications using CST Microwave Studio, ensuring enhanced radiation efficiency and reduced SAR.
- Fabricated and validated a flexible antenna prototype on Kapton substrate; achieved –16dB return loss and stable radiation patterns under bending, ensuring robust performance for wearable IoT health monitoring.

- Intelligent Fog Detection and Vehicle Alert System *Project Report*

- Developed an intelligent camera system on Raspberry Pi using AI (TensorFlow) and computer vision (OpenCV) to detect vehicles in real-time, significantly improving driving safety in foggy conditions.
- Implemented Dark Channel Prior for real-time fog removal, enhancing visual clarity and improving vehicle detection accuracy for safer navigation.
- Engineered a smart alert logic to estimate collision risk and suggest alternate routes, enabling faster driver response in low-visibility scenarios.

Certifications

- **Oracle:** Java Fundamentals, Java Foundation, Java Programming *Certificate*
- **AWS:** Certified Solutions Architect - Associate *Certificate*
- **NPTEL:** Introduction to Internet of Things *Certificate*
- **NPTEL:** Sensors and Actuators *Certificate*

Technical Skills

Languages: Java, Python, MySQL
Microsoft Office : Word, Excel, PowerPoint
Google Workspace : Docs, Sheets, Slides
Software: CST(Computer Simulation Technology), Matlab